MUSTAPHA ISMAEL OMAR

Pretoria, South Africa|081 533 7351 |mustaphaismaelomar786@gmail.com |
LinkedIn: linkedin.com/in/mustapha-ismael-omar-2a2b11318/ | **GitHub:** https://github.com/Musti-Omar-97993

PROFESSIONAL SUMMARY

Data Engineer and BComp Honours Data Science candidate with enterprise experience at Vodacom, Atos, Siemens, and Orange Business Solutions. Experienced in cloud data infrastructure, ETL/ELT pipelines, SQL, IoT systems, machine learning, and full-stack development. Strong practical experience with AWS data services and enterprise technology environments, complemented by international multidisciplinary project experience and technical leadership.

TECHNICAL & PROFESSIONAL SKILLS

- Cloud & Data Engineering: AWS (Glue, S3, Athena, Redshift, Managed Airflow), Data Lakehouse Architecture, ETL/ELT Pipelines, SQL Server, MySQL, Oracle, MongoDB, Talend
- Data Science & Analytics: Pandas, NumPy, Scikit-learn, TensorFlow, Keras, Power BI, Tableau, Advanced Excel
- Programming Languages: Python, SQL, C#, JavaScript, HTML5/CSS3
- Development Tools & DevOps: Git/GitHub, Docker, FastAPI, Firebase, VS Code, Visual Studio, JupyterLab, Google Colab
- IoT & Networking: MQTT, Node-RED, Arduino IDE, Cisco Packet Tracer, Network Infrastructure & Troubleshooting, LAN Cabling, AP Installation
- Enterprise IT & Security: Hardware Provisioning, OS Imaging, User Onboarding, PKI/Certificate Management, Microsoft Intune, ServiceNow, MFA Configuration
- **Professional & Soft Skills:** Cross-Functional Team Leadership, Project Management, Stakeholder Communication, Financial Planning & Budgeting, Public Speaking, Problem Solving, Agile Collaboration
- Languages: English (Native / Full Professional), Afrikaans (Basic / Elementary Proficiency)

EDUCATION

BComp Honours in Data Science (Integrated Programme) | Belgium Campus ITversity, Pretoria
2023 – Early 2027 | Graduation: Early 2027

National Senior Certificate (Matric) | Laudium Secondary School Class of 2021

WORK EXPERIENCE

Atos South Africa: *Junior Engineer (Intern*) | March 2026 – Present
Currently completing an internship with Atos South Africa, gaining practical experience across network engineering, technical support, and data engineering through assignments with Atos' enterprise clients.

Vodacom: *Junior Data Engineer* | July 2026 – Present

- Engineered backend data solutions for the high-volume VodaPay enterprise application.
- Perform data extraction, transformation, and processing activities and develop SQL queries for data requirements.
- Developed and maintained robust ETL/ELT pipelines, implementing rigorous data quality and validation checks.
- Architected scalable cloud data lakehouse solutions utilizing AWS Glue, S3, Athena, Airflow, and Redshift to optimize data ingestion, transformation, and analytics.

Siemens: *Junior Network & Support Engineer* | March 2026 – July 2026

- Executed hardware provisioning, OS imaging, and compliance configurations across enterprise networks.
- Provided desk-side and remote support for software, network connectivity, authentication, and user access.
- Troubleshot technical incidents involving hardware, connectivity, and system performance.
- Assisted with network infrastructure maintenance, hardware installations, and LAN cabling.

Orange Business Solutions: *Junior Support Engineer* | March 2026 – July 2026

- Provided first- and second-line support for technical incidents and service requests.
- Assisted with device imaging, user setup, and technical configuration.
- Maintained technical documentation and records of infrastructure changes and support activities.

PROJECTS & INTERNATIONAL EXPERIENCE

Global Acting in IT Minor 2025: International Academic Exchange & Environmental IT Project

- Successfully completed a study exchange across five international universities: Fontys (Netherlands), University of Lleida (Spain), FH Oberösterreich (Austria), Hasselt University (Belgium), and Belgium Campus (South Africa).
- This program specialized in Immersive Tech, UX Design, IoT, AI, and Data Science.
- The project was to design and build a device, and dashboard to monitor the water quality levels for dams in South Africa. And create a navigation application for boaters to navigate around obstacles that could be in the dam, such as hyacinth mats.
- My contribution – Architected the IoT infrastructure by building and calibrating sensors, configuring gateways, and deploying MQTT brokers via The Things Network. Developed machine learning classification models leveraging environmental data, and engineered the supporting SQL databases and interactive front-end dashboards.

Penn State 2025: International Engineering Project Smart Hydroponics System with Dashboard & Educational Platform

- The project aim was to develop a smart hydroponics system with real-time monitoring via a Node-RED dashboard and an educational website.

- My contribution – Programmed automated sensor control logic for precision water and nutrient distribution. Developed an interactive, full-stack educational platform to teach sustainable hydroponics to primary school learners.

BusIT Week 2024 – Smart Hydroponics Sensor Integration

- The aim of this project was to automate a hydroponics systems using sensors to monitor water level and pH in a hydroponics control system.
- My contribution to this project was setting up and calibrating sensors, as well as building the dashboard.

Role-Based POS & General Management System

- Architected and deployed a custom C# Windows Forms POS and management system, integrating secure user authentication, financial tracking, and SQL-based stock management.
- This was a side project that I developed independently from start to finish.

For more projects you can check out my github.

LEADERSHIP EXPERIENCE

SRC Head of Finance (2024) *Belgium Campus ITversity, Pretoria*

- Elected to manage budget planning, financial reporting, and the allocation of student funds, while coordinating with student committees to ensure transparent and efficient use of financial resources.
- Represented Belgium Campus ITversity during Open Days by conducting campus tours and demonstrating student projects to prospective students, parents, and visitors.
- Engaged with visitors by providing information about the campus, student experience, and academic programmes, developing strong leadership, communication, public speaking, and organisational skills.

Project Lead*: International Engineering Projects (2025)*

- Led the IoT and Data components of the Global Acting in IT Minor environmental technology project, coordinating the development and integration of sensor systems, IoT infrastructure, databases, and machine learning components.
- Served as a rotating team lead during the Penn State International Engineering Project, taking responsibility for team coordination, progress monitoring, and project deliverables during assigned two-week leadership periods.

HOBBIES & INTERESTS

Sustainable Farming & Agriculture Technology | Artificial Intelligence & Emerging Technologies | IoT & Smart Systems | Technology & DIY Projects | Vintage Car Restoration | Soccer | Hiking & Outdoor Activities | Exploring New Places & Cultures

For additional information please visit my online CV using the link below.

https://cv-website-phvm.onrender.com